Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 1 of 10 · find part e in this mark scheme extract

1(a)	**The only correct answer is C** A is not correct because Boolean only has values True and False B is not correct because it only allows for one character D is not correct because it is used for mixed text and numeric data		(1)

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 2 of 10 · find part e in this mark scheme extract

1(b)	(an error that) produces incorrect /unintended output (1)	Accept an example of an explained logic error	(1)

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 3 of 10 · find part e in this mark scheme extract

1(d)	Any one from: 1. explain how the program works (1) 2. improve readability of the program (1) 3. make code more understandable (1)		(1)

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 4 of 10 · find part e in this mark scheme extract

1(e)	1. Global variables created in the main program / can be accessed from any part of the program / are in scope throughout the program (1) 2. Local variables are declared/created inside a function/blocked code (indented, brackets) and are only accessible there / are only in scope within the function in which they're declared (1)		(2)

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 5 of 10 · find part e in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 6 of 10 · find part e in this mark scheme extract

<table>
<tr>
<td>1(g)</td>
<td>For up to two tests
Award 1 mark for an appropriate validation test and 1 mark for an item of test data that would fail only the given test

<table>
<tr><th>Validation test</th><th>Erroneous data</th></tr>
<tr><td>Are the first two characters letters/upper case letters?</td><td>1234CCC</td></tr>
<tr><td>Are the third- and fourth-characters numerical digits?</td><td>CCCCBBB</td></tr>
<tr><td>Are the last three characters letters/upper case letters?</td><td>CC12345</td></tr>
</table>
</td>
<td>Each test should target a different feature of one of the requirements

Exactly 7 characters (in the stem)
First two characters are letters / upper case letters
Characters 3 & 4 are numeric
Last three characters are letters / upper case letters

</td>
<td>(4)</td>
</tr>
</table>

Question	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 7 of 10 · find part e in this mark scheme extract

1(c)	**C#** 1. Semicolon at the end of line 9 [maxValue=5;] (1) 2. Consistent capitalisation of 'count' on lines 10, 11, 13 and 14 (1) 3. constantValue needs to be defined and given the value 7 / replaced by value 7 (1) **Java** 1. Semicolon at the end of line 8 [maxValue = 5;] (1) 2. Consistent capitalisation of 'count' on lines 7, 9, 11 and 12 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1) **Python** 1. Colon needed at end of line 7 [count, maxValue):] (1) 2. Consistent capitalisation of 'count' on line 3, 7, 8 and 9 (1) 3. constantValue needs to be defined and given the value 7 / replaced by 7 (1)	Candidates are required to open the file Q01c in the code editor. Amended code should be saved as Q01cFINISHED. Line numbers may have been changed by corrections to code (e.g. insertion of constantValue definition)	(3)

Question	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 8 of 10 · find part e in this mark scheme extract

<table>
<tr>
<td>1(f)</td>
<td>Award 1 mark for each correct pair of relational operators and 1 mark for correct Boolean operator for each condition.

<table>
<tr><th>Condition</th><th>Coding</th></tr>
<tr><td rowspan="2">Shop income more than £5000 or assistant sold at least 10 pairs</td><td>shopIncome > 5000; assistantSales >= 10 (1)</td></tr>
<tr><td>correct OR operator (1)</td></tr>
<tr><td rowspan="2">Shop income at least £2000 and assistant sold at least 5 pairs</td><td>shopIncome >= 2000; assistantSales >= 5 (1)</td></tr>
<tr><td>correct AND operator (1)</td></tr>
</table>

Accept >9, >1999 and >4 for respective >= comparisons</td>
<td>Candidates are required to open the file Q01f in the code editor.
Amended code should be saved as Q01fFINISHED.
Do not penalise candidates who attempt more than the stated requirements.
Do not award Boolean operator mark for single | or & as these are not correct.</td>
<td>(4)</td>
</tr>
</table>

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 9 of 10 · find part e in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 1 of 4 · Question 1 part e · 2 marks

2019 JUN · Paper 02 · extract 10 of 10 · find part e in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 2 of 4 · Question 2 part a.iii · 1 mark

2023 JUN · Paper 02 · extract 1 of 4 · find part a.iii in this mark scheme extract

2(a)(i)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • The code in the library has already been debugged/tested/optimised / library code less likely to have errors (1) • May include specialised/proprietary functions (1) • Library subprograms can be imported into code when needed / don't need to write code themselves (to save a programmer time/work) (1) • making it more readable / makes code easier to understand (Code files become shorter)	Allow characteristics of a specific library (math, random), if it can be attributed to unique bullets Don't allow quicker/easier/better without a reason being given Accept answers about subprograms as candidate may have written these themselves Line numbers are for guidance – mark answer as a whole	**2**
2(a)(ii)	A3	**The only correct answer is A** ***B*** *is not correct because iteration is not callable and does not return a result.* ***C*** *is not correct because a procedure, while callable, does not return a result.* ***D*** *is not correct because selection is not callable and does not return a result.*		**1**
2(a)(iii)	A4	Award **one** mark for: • A variable that exists/is accessible only in the subprogram/scope in which it is created (1)		**1**

Type MS · Local and global variables

Answer 2 of 4 · Question 2 part a.iii · 1 mark

2023 JUN · Paper 02 · extract 2 of 4 · find part a.iii in this mark scheme extract

Type MS · Local and global variables

Answer 2 of 4 · Question 2 part a.iii · 1 mark

2023 JUN · Paper 02 · extract 3 of 4 · find part a.iii in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(b)(i)</td><td>B1
B2
B3</td><td>Award one mark for each correct row.
<table><tr><th></th><th>Adult</th><th>Children</th></tr><tr><td>Normal</td><td>1/2/3/4</td><td>1/2/3/4/5/6</td></tr><tr><td>Boundary</td><td>1/4</td><td>1/6</td></tr><tr><td>Erroneous</td><td>Any value <1 or >4</td><td>Any value <1 or >6</td></tr></table></td><td>Accept all valid values for normal and boundary
Do not accept written numbers (two) as question asks for ‘numeric data’ – this is testing range, not data types.</td><td>3</td></tr>
<tr><td rowspan="5">2(b)(ii)</td><td colspan="2">Award one mark for each of:</td><td rowspan="5">Runtime error (B7) could be fixed before logic error (B6)
Applies to all languages
Accept < for <=
Allow /-1000 as alternative to B6
/1000 could happen after initial location e.g. print(“Order”, weightNeed/1000, “kilograms)</td><td rowspan="5">4</td></tr>
<tr><td>B4</td><td>Change =+= to +</td></tr>
<tr><td>B5</td><td>Change > to <=</td></tr>
<tr><td>B6</td><td>Change order of subtraction to: requiredWeight - inStock</td></tr>
<tr><td>B7</td><td>Change 0 to 1000</td></tr>
</table>

Type MS · Local and global variables

Answer 2 of 4 · Question 2 part a.iii · 1 mark

2023 JUN · Paper 02 · extract 4 of 4 · find part a.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 1 of 9 · find part a.i in this mark scheme extract

4(a)(i)	A1	Award **one** mark for: • (At the highest level) in the main scope/program/file (1)	Java and C#: allow answers that indicate inside namespace e.g. 'before main class'	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 2 of 9 · find part a.i in this mark scheme extract

4(a)(ii)	A2	Award **one** mark for: • Only from within the scope/subprogram/code block in which it is created/declared (1)		**1**

Question	mp	Answer	Additional guidance	Mark

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 3 of 9 · find part a.i in this mark scheme extract

4(b)	B1 B2	Award **two** marks for a linked explanation such as: • **Figure 4** executes fewer comparisons/tests (1), because (the use of nested ifs/else ifs/ means that) as soon as a test evaluates to true, all the remaining tests are skipped (1) • **Figure 5** executes more comparison/tests (1), because every if statement must be executed, even if any test before has already evaluated to true (1)		**2**

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 4 of 9 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 5 of 9 · find part a.i in this mark scheme extract

4(c)		Award **one** mark for each:		
	C1	Subprogram must attempt to use both pString and pNum to generate Key (1)	Award attempted use, even if not correct	
	C2	1D indexing used to access chars in string (1)	Can be anywhere in code (main or subprogram) Any method e.g. • pString[0]	
	C3	2 pairs of separated chars obtained (1)	First 2 **and** last 2 characters obtained from pString Any method Can be anywhere in code • pString[0] + pString[1] • pString[2] + pString[3] • pString[0:2] • pString[2:4]	
	C4	Key concatenated correctly (1)	Key must be correct Can be anywhere in code	
	C5	Subprogram must return their key to the calling line (1)	Do not award return without a value • Value must be variable/value intended to be key	
	C6	Complete line to call new subprogram (1)	Must be fully correct `myNewKey = genNewKey(myString, myNumber)`	**6**

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 6 of 9 · find part a.i in this mark scheme extract

C#

```
// Q04cFINISHED

namespace Q04cFINISHED
{
    class Q04cFINISHED
    {
        static String genNewKey (String pString, int pNum)
        {
            String newKey;

            // Write your code below this line
            newKey = pString.Substring (0, 2);            // First two chars
            newKey = newKey + pNum.ToString();            // Number
            newKey = newKey + pString.Substring (2, 2);   // Last two chars

            return (newKey);
        }

        static void Main(string[] args)
        {
            String myNewKey = "";
            int myNumber = 0;
            String myString = "";

            Console.WriteLine ("Enter a string:");
            myString = Console.ReadLine();
            Console.WriteLine ("Enter a whole number: ");
            myNumber = Convert.ToInt32(Console.ReadLine());

            if (myString.Length != 4)
            {
                Console.WriteLine ("String must be four characters");
            }
            else
            {
                Console.WriteLine ("Original: " + myString +
                                   " " + myNumber.ToString());

                // Complete the call to the subprogram
                myNewKey = genNewKey (myString, myNumber);
                Console.WriteLine ("New: " + myNewKey);
            }
        }
    }
}
```

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 7 of 9 · find part a.i in this mark scheme extract

Java

```java
// Q04cFINISHED

import java.util.Scanner;

public class Q04cFINISHED
{
    static String genNewKey (String pString, int pNum)
    {
        String newKey;

        // Write your code below this line
        newKey = pString.substring(0,2);  // First two chars
        newKey = newKey + String.valueOf (pNum);              // Number
        newKey = newKey + pString.substring (2, 4);   // Last two chars

        return (newKey);
    }

    public static void main(String[] args)
    {
        String myNewKey = "";
        int myNumber = 0;
        String myString = "";
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a string:");
        myString = myKeyboard.nextLine();
        System.out.println ("Enter a whole number: ");
        myNumber = myKeyboard.nextInt();

        if (myString.length() != 4)
        {
            System.out.println ("String must be four characters");
        }
        else
        {
            System.out.println ("Original: " + myString +
                                " " + String.valueOf(myNumber));

            // Complete the call to the subprogram
            myNewKey = genNewKey (myString, myNumber);
            System.out.println ("New: " + myNewKey);
        }
    }
}

```

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 8 of 9 · find part a.i in this mark scheme extract

Python

```
# Q04cFINISHED

def genNewKey (pString, pNum):
    newKey = ""         # Make new word here

    # Write your code below this line
    newKey = pString[0:2]               # First two chars
    newKey = newKey + str (pNum)        # Number
    newKey = newKey + pString[2:4]      # Last two chars
    return (newKey)

myString = input ("Enter a string: ")
myNumber = int (input ("Enter a whole number: "))

if (len (myString) != 4):
    print ("String must be four characters")
else:
    print ("Original: ", myString, myNumber)

    # Complete the call to the subprogram
    myNewKey = genNewKey (myString, myNumber)
    print ("New word:", myNewKey)

```

Type MS · Local and global variables

Answer 3 of 4 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 · extract 9 of 9 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 1 of 9 · find part a.ii in this mark scheme extract

4(a)(i)	A1	Award **one** mark for: • (At the highest level) in the main scope/program/file (1)	Java and C#: allow answers that indicate inside namespace e.g. 'before main class'	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 2 of 9 · find part a.ii in this mark scheme extract

4(a)(ii)	A2	Award **one** mark for: • Only from within the scope/subprogram/code block in which it is created/declared (1)		**1**

Question	mp	Answer	Additional guidance	Mark

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 3 of 9 · find part a.ii in this mark scheme extract

4(b)	B1 B2	Award **two** marks for a linked explanation such as: • **Figure 4** executes fewer comparisons/tests (1), because (the use of nested ifs/else ifs/ means that) as soon as a test evaluates to true, all the remaining tests are skipped (1) • **Figure 5** executes more comparison/tests (1), because every if statement must be executed, even if any test before has already evaluated to true (1)		**2**

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 4 of 9 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 5 of 9 · find part a.ii in this mark scheme extract

4(c)		Award **one** mark for each:		
	C1	Subprogram must attempt to use both pString and pNum to generate Key (1)	Award attempted use, even if not correct	
	C2	1D indexing used to access chars in string (1)	Can be anywhere in code (main or subprogram) Any method e.g. • pString[0]	
	C3	2 pairs of separated chars obtained (1)	First 2 **and** last 2 characters obtained from pString Any method Can be anywhere in code • pString[0] + pString[1] • pString[2] + pString[3] • pString[0:2] • pString[2:4]	
	C4	Key concatenated correctly (1)	Key must be correct Can be anywhere in code	
	C5	Subprogram must return their key to the calling line (1)	Do not award return without a value • Value must be variable/value intended to be key	
	C6	Complete line to call new subprogram (1)	Must be fully correct `myNewKey = genNewKey(myString, myNumber)`	**6**

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 6 of 9 · find part a.ii in this mark scheme extract

C#

```
// Q04cFINISHED

namespace Q04cFINISHED
{
    class Q04cFINISHED
    {
        static String genNewKey (String pString, int pNum)
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            String newKey;

            // Write your code below this line
            newKey = pString.Substring (0, 2);            // First two chars
            newKey = newKey + pNum.ToString();            // Number
            newKey = newKey + pString.Substring (2, 2);   // Last two chars

            return (newKey);
        }

        static void Main(string[] args)
        {
            String myNewKey = "";
            int myNumber = 0;
            String myString = "";

            Console.WriteLine ("Enter a string:");
            myString = Console.ReadLine();
            Console.WriteLine ("Enter a whole number: ");
            myNumber = Convert.ToInt32(Console.ReadLine());

            if (myString.Length != 4)
            {
                Console.WriteLine ("String must be four characters");
            }
            else
            {
                Console.WriteLine ("Original: " + myString +
                                   " " + myNumber.ToString());

                // Complete the call to the subprogram
                myNewKey = genNewKey (myString, myNumber);
                Console.WriteLine ("New: " + myNewKey);
            }
        }
    }
}

```

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 7 of 9 · find part a.ii in this mark scheme extract

Java

```java
// Q04cFINISHED

import java.util.Scanner;

public class Q04cFINISHED
{
    static String genNewKey (String pString, int pNum)
    {
        String newKey;

        // Write your code below this line
        newKey = pString.substring(0,2);  // First two chars
        newKey = newKey + String.valueOf (pNum);              // Number
        newKey = newKey + pString.substring (2, 4);   // Last two chars

        return (newKey);
    }

    public static void main(String[] args)
    {
        String myNewKey = "";
        int myNumber = 0;
        String myString = "";
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a string:");
        myString = myKeyboard.nextLine();
        System.out.println ("Enter a whole number: ");
        myNumber = myKeyboard.nextInt();

        if (myString.length() != 4)
        {
            System.out.println ("String must be four characters");
        }
        else
        {
            System.out.println ("Original: " + myString +
                                " " + String.valueOf(myNumber));

            // Complete the call to the subprogram
            myNewKey = genNewKey (myString, myNumber);
            System.out.println ("New: " + myNewKey);
        }
    }
}

```

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 8 of 9 · find part a.ii in this mark scheme extract

Python

```
# Q04cFINISHED

def genNewKey (pString, pNum):
    newKey = ""         # Make new word here

    # Write your code below this line
    newKey = pString[0:2]               # First two chars
    newKey = newKey + str (pNum)        # Number
    newKey = newKey + pString[2:4]      # Last two chars
    return (newKey)

myString = input ("Enter a string: ")
myNumber = int (input ("Enter a whole number: "))

if (len (myString) != 4):
    print ("String must be four characters")
else:
    print ("Original: ", myString, myNumber)

    # Complete the call to the subprogram
    myNewKey = genNewKey (myString, myNumber)
    print ("New word:", myNewKey)

```

Type MS · Local and global variables

Answer 4 of 4 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 02 · extract 9 of 9 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark